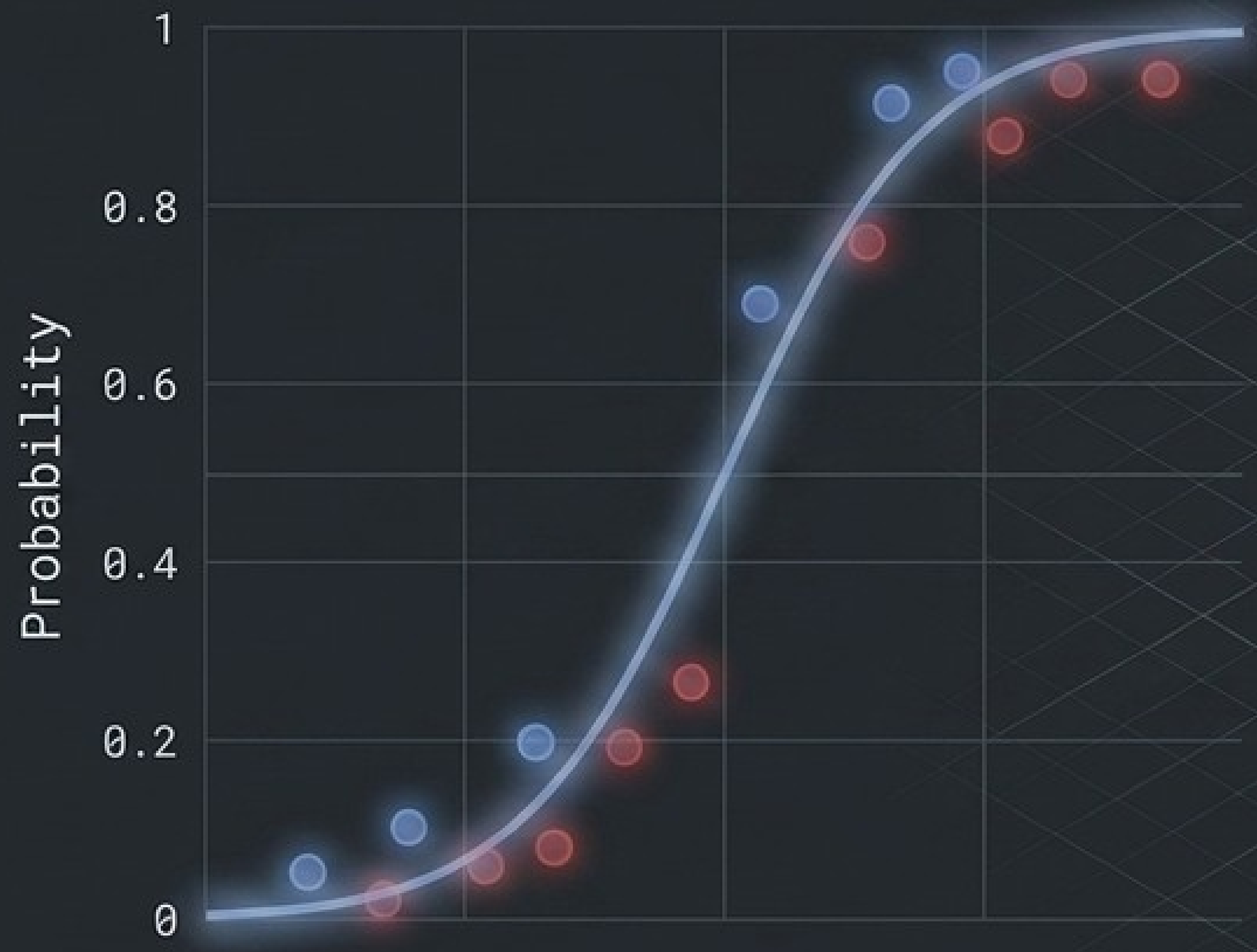
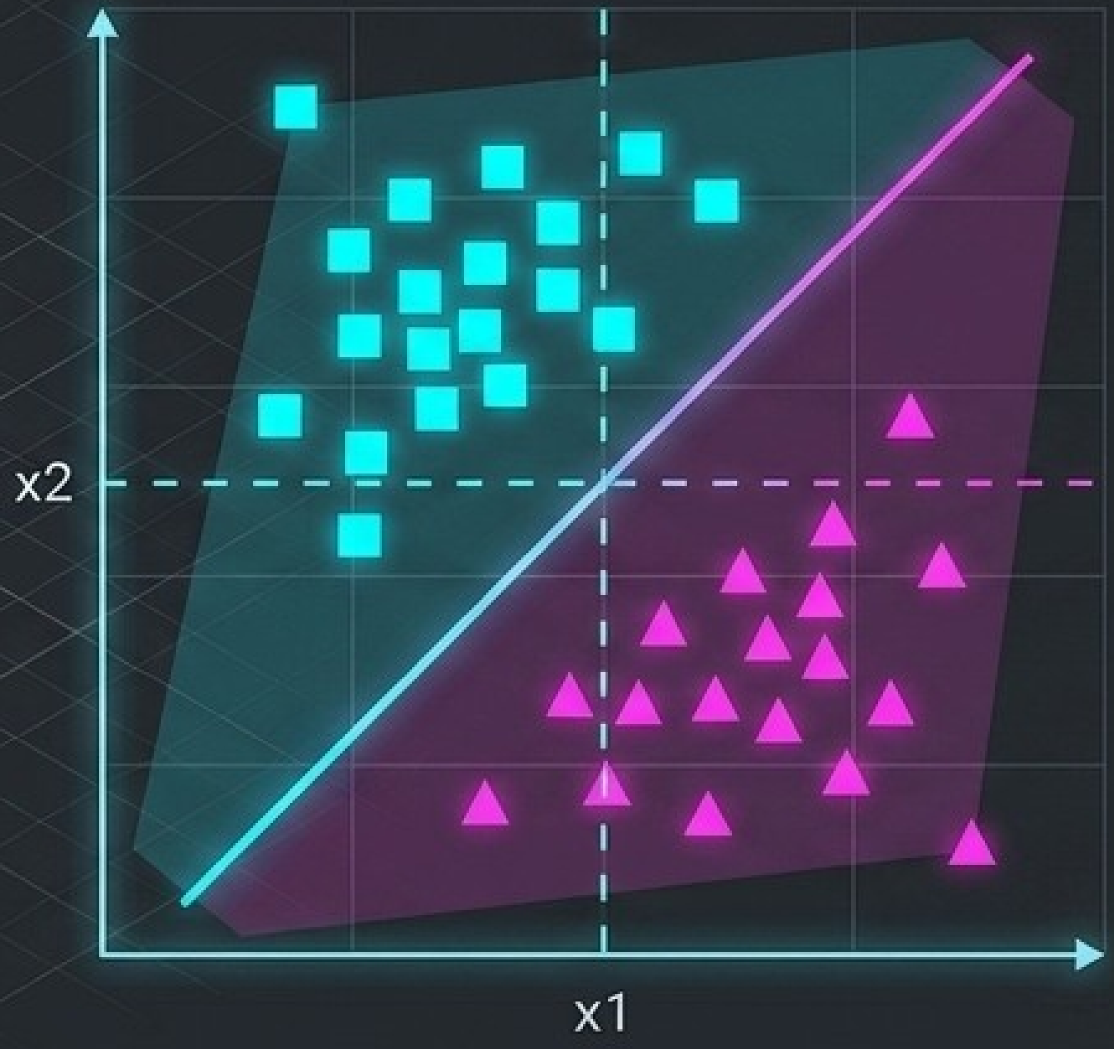
Beyond Statistics: The Geometry of Classification



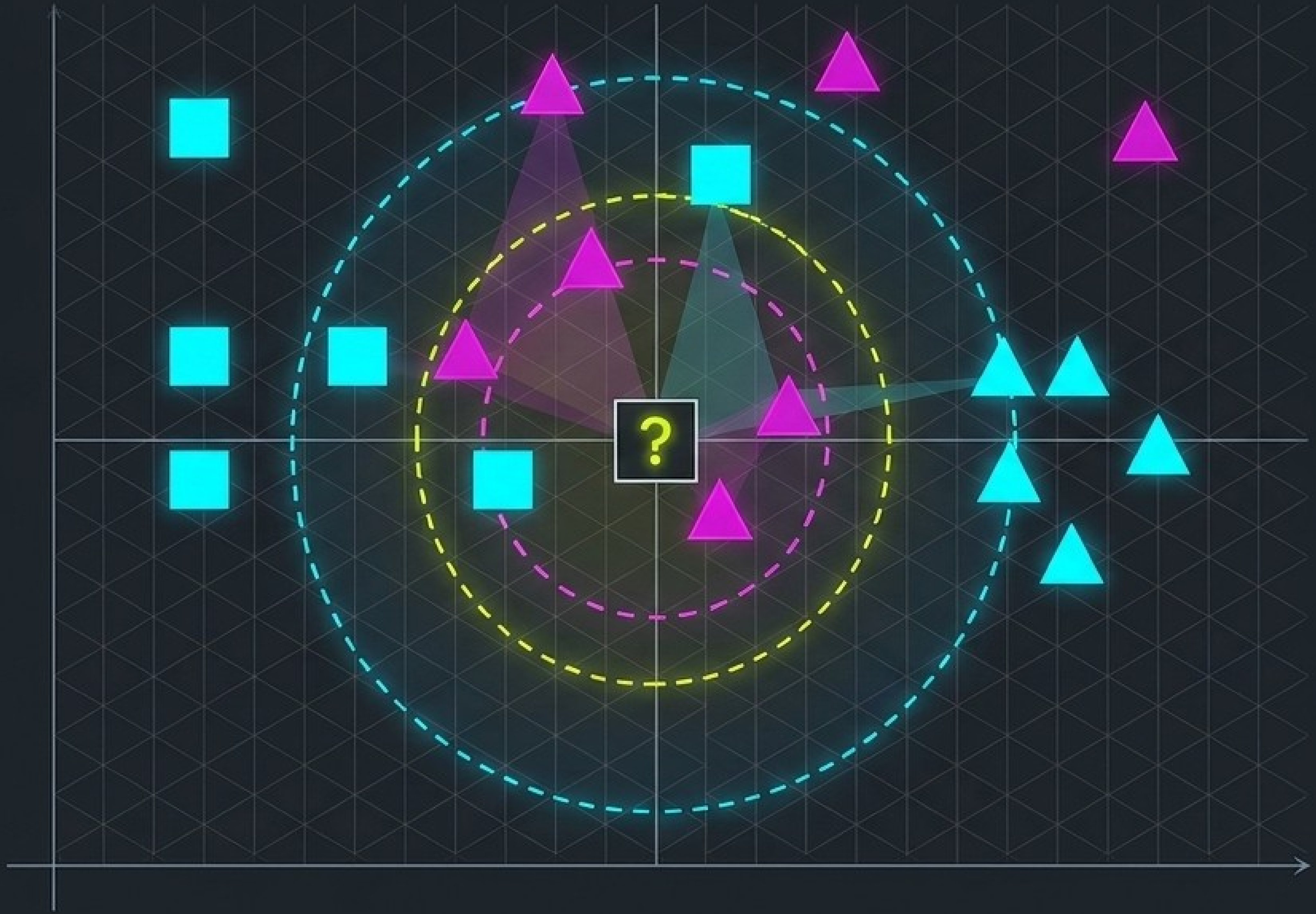


Previous Paradigm: Statistical models fitting curves to predict probabilities.

New Paradigm: Instance-based mapping and rigid spatial boundaries.

Focus Areas: K-Nearest Neighbors (KNN) & Support Vector Machines (SVM).

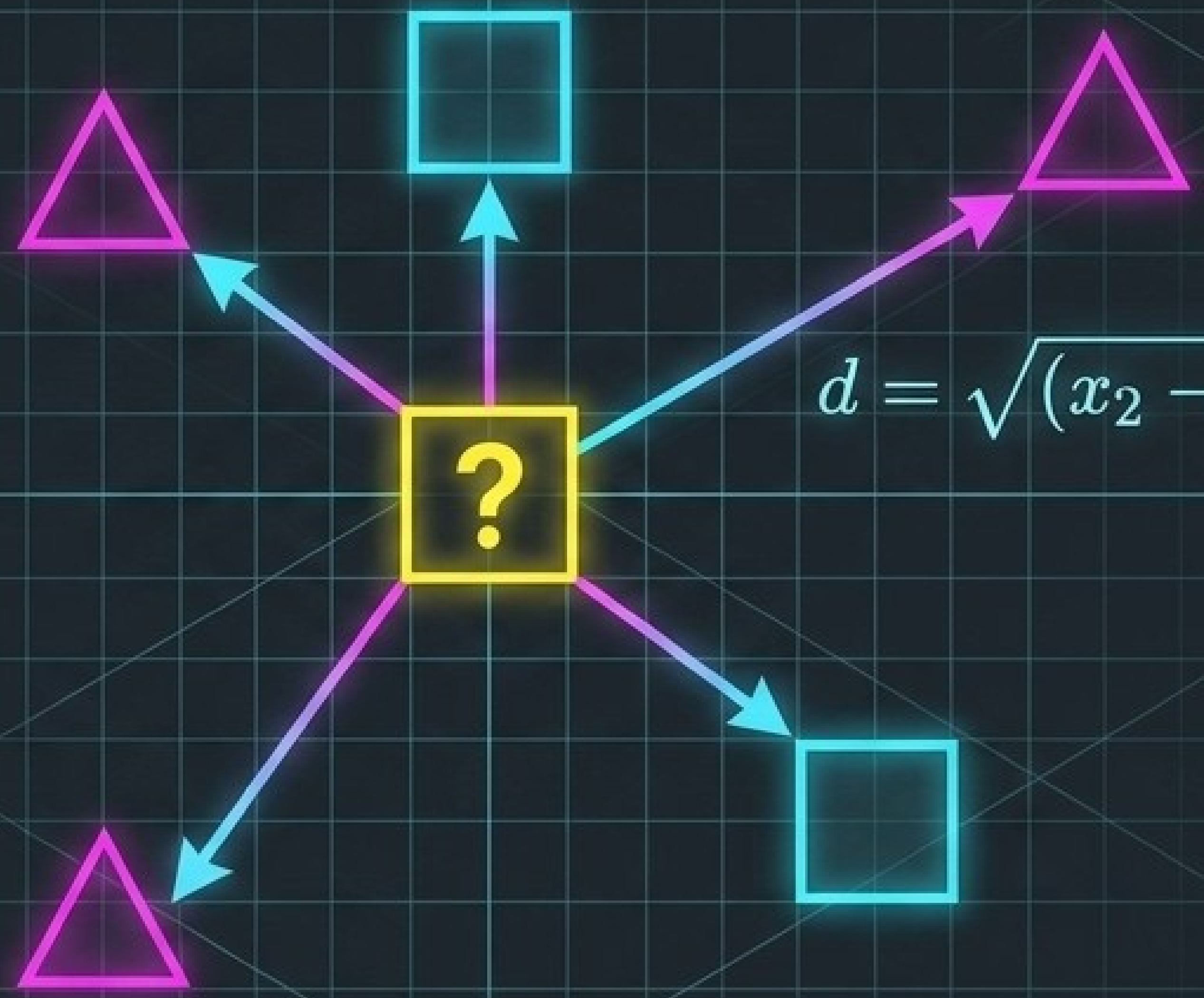
The Lazy Learner Approach



- No Training Phase: Builds no internal model; simply memorizes raw data.
- Query-Time Action: Computation deferred until a prediction is requested.
- Non-Parametric: Zero underlying mathematical assumptions about data.

Measuring Space and Majority Rule

- Distance Metrics: Proximity defined by spatial measurement (e.g., Euclidean).
- Neighborhood Size (K): Dictates exactly how many adjacent points get a vote.
- Majority Voting: The most frequent class within the radius dictates prediction.



$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

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3 Magenta, 2 Cyan
-> Predicted Magenta
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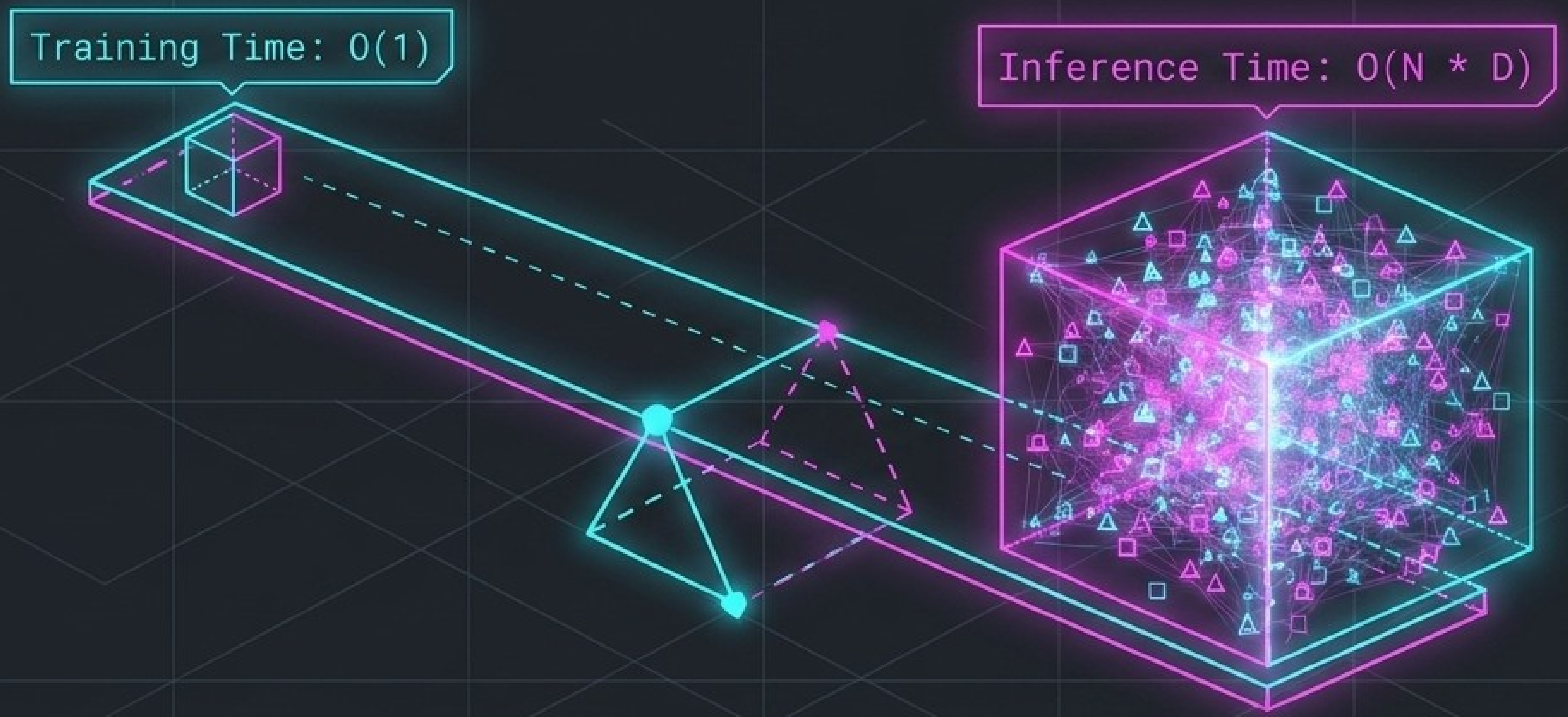
The Computational Cost of Laziness

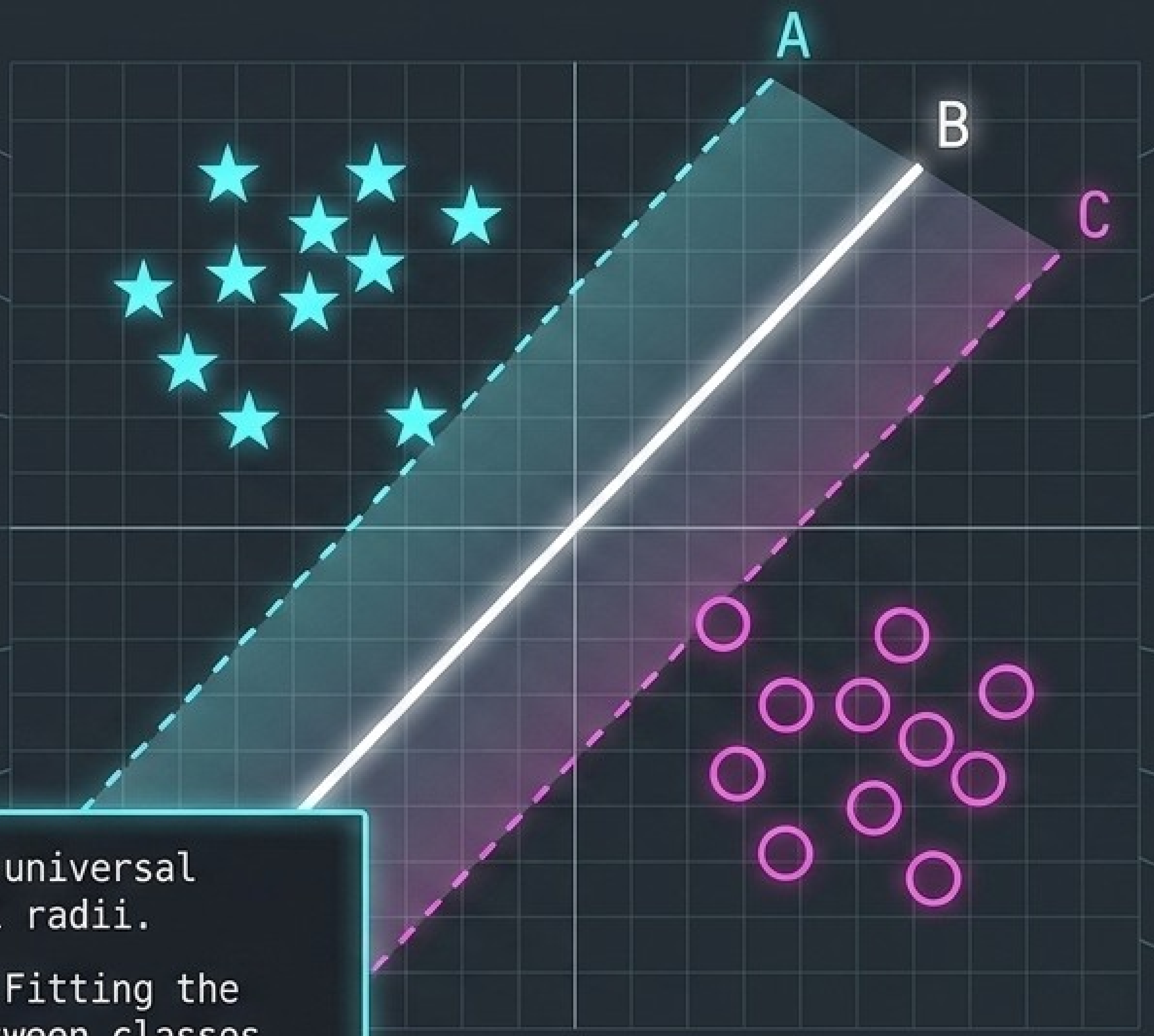


Instantaneous Training:
Adding new data is immediate and cost-free.

Expensive Prediction:
Requires calculating distance to every historical point.

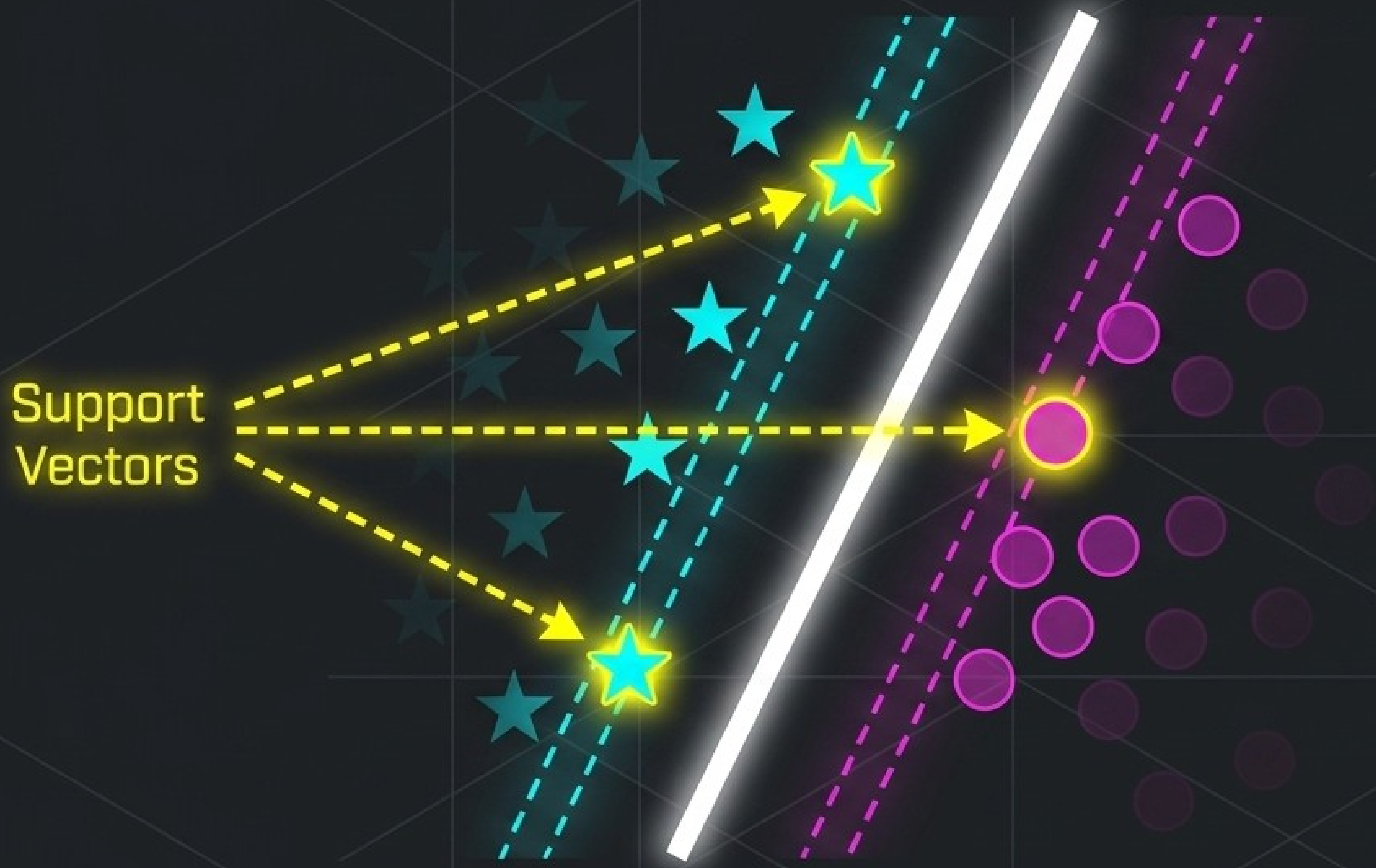
Memory Bound:
Entire massive dataset must remain loaded in RAM.

Paving the Widest Possible Street



- Global Boundary: A singular, universal dividing plane replaces local radii.
- Large Margin Classification: Fitting the maximum possible distance between classes.
- Robust Generalization: Maximizing the margin ensures reliable future predictions.

The Edge of the Street

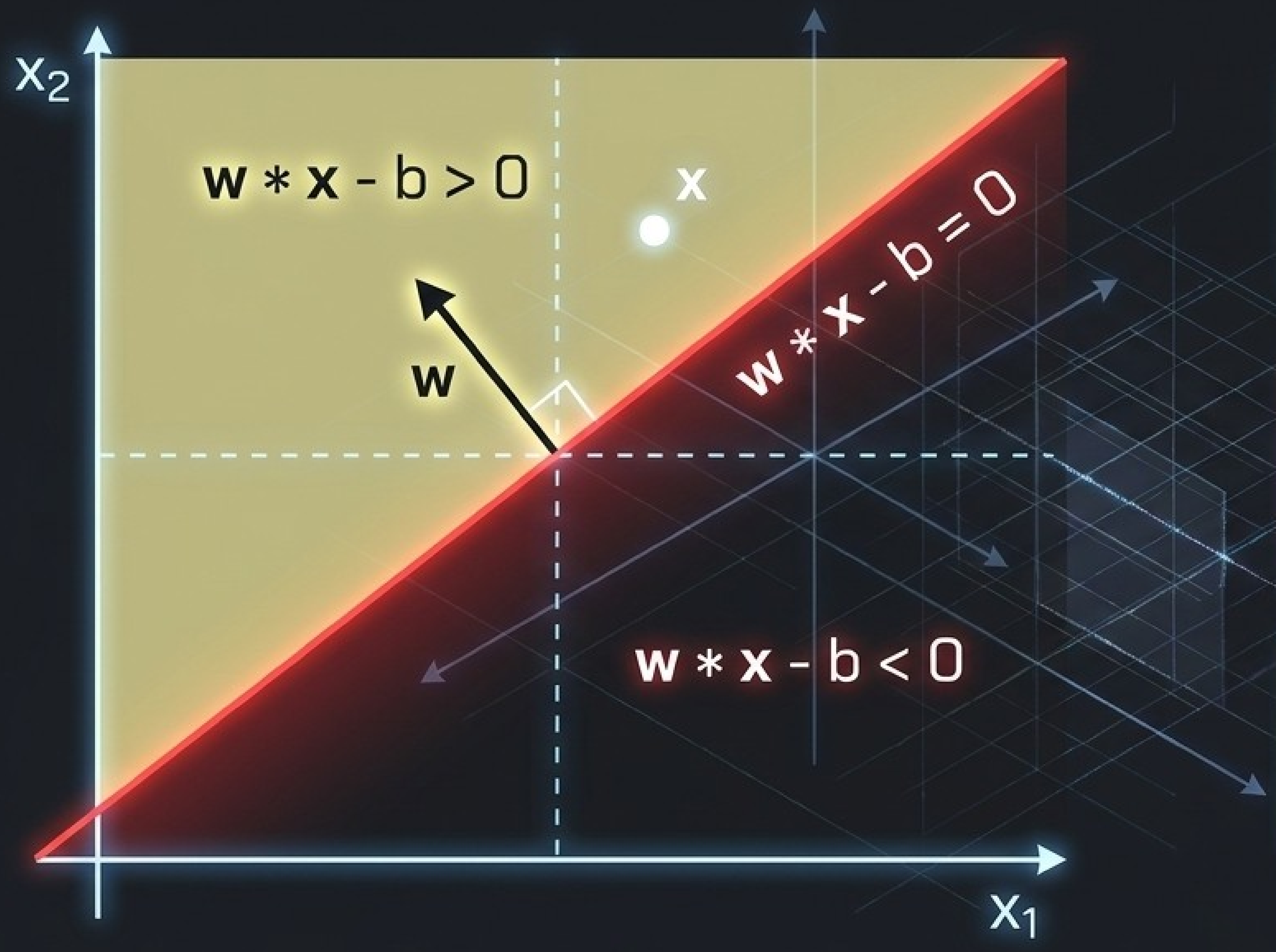


Critical Instances:
The boundary is dictated only by points directly touching the margin.

Ignore the Rest:
Modifying or deleting non-support vectors changes absolutely nothing.

High Efficiency:
The model is defined by a microscopic fraction of the dataset.

Formulating the Spatial Boundary

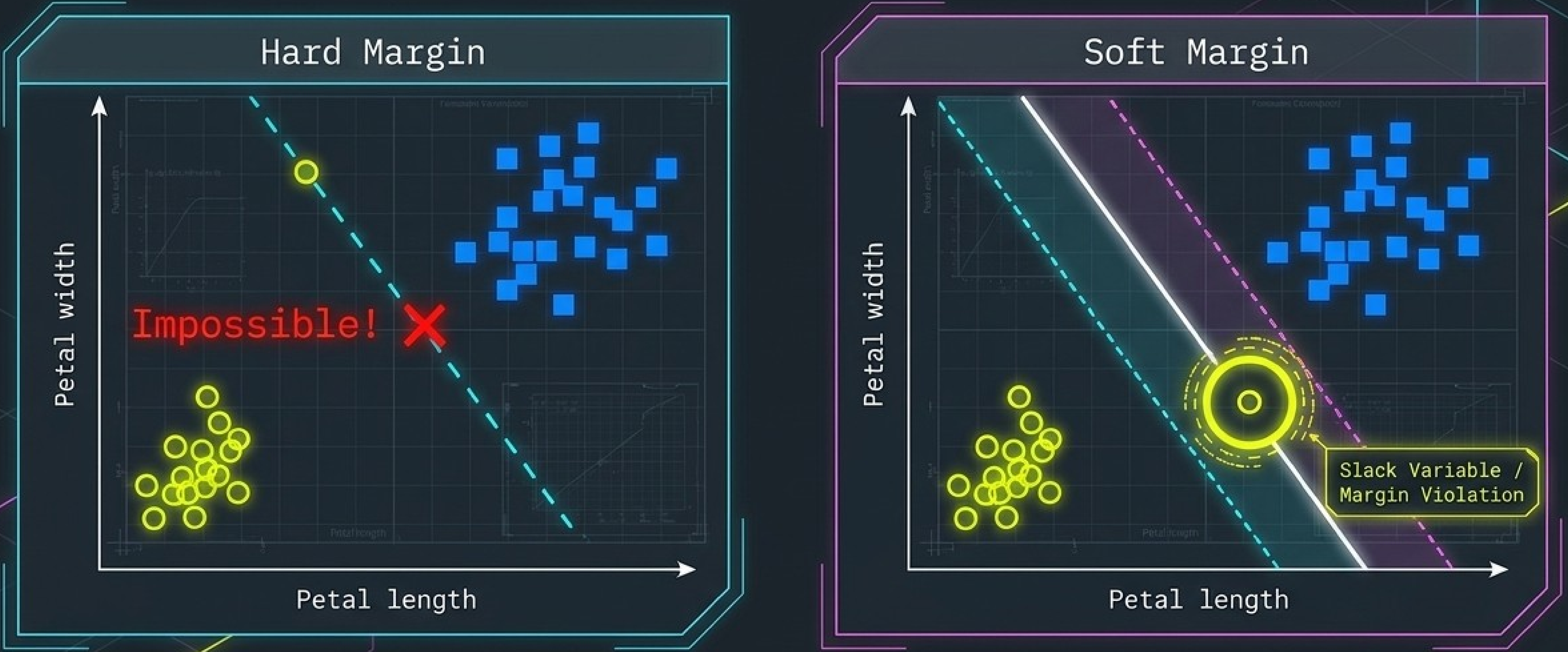


The Separator: Boundary rigidly defined by weight vector **w** orthogonal to the hyperplane.

Positive Space:
If dot product yields $\mathbf{w} * \mathbf{x} - b > 0$, **predict positive class (+1).**

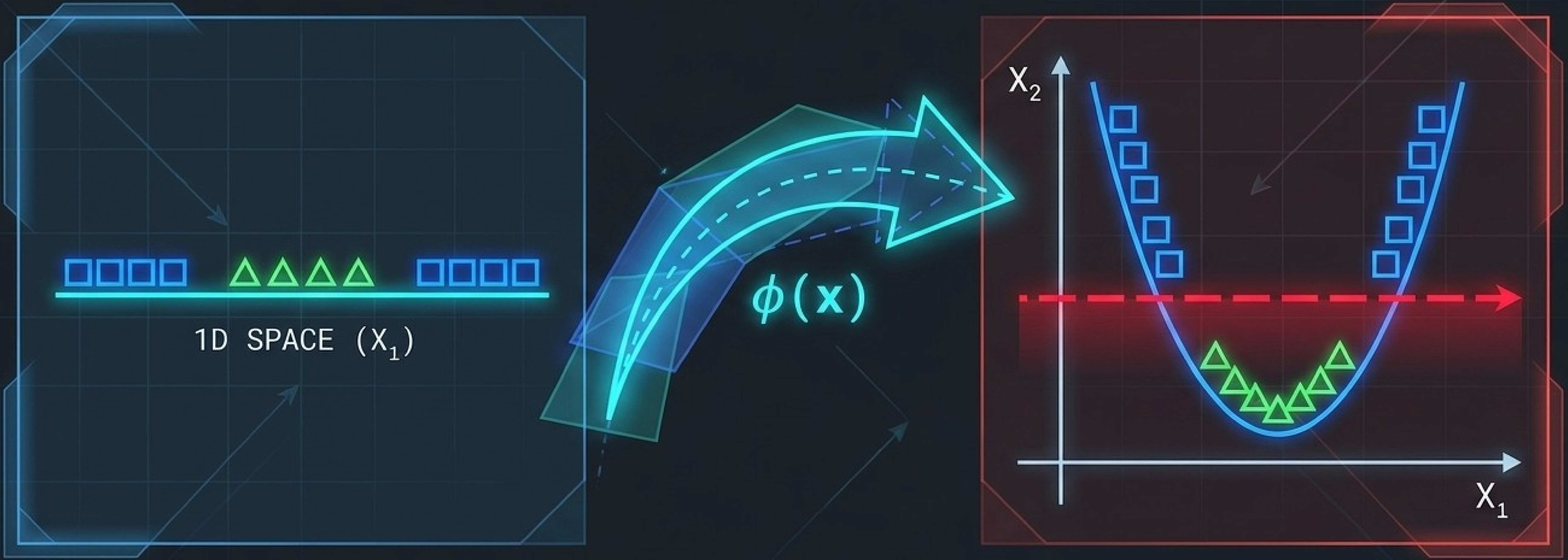
Negative Space:
If dot product yields $\mathbf{w} * \mathbf{x} - b < 0$, **predict negative class (-1).**

The Reality of Noisy Data



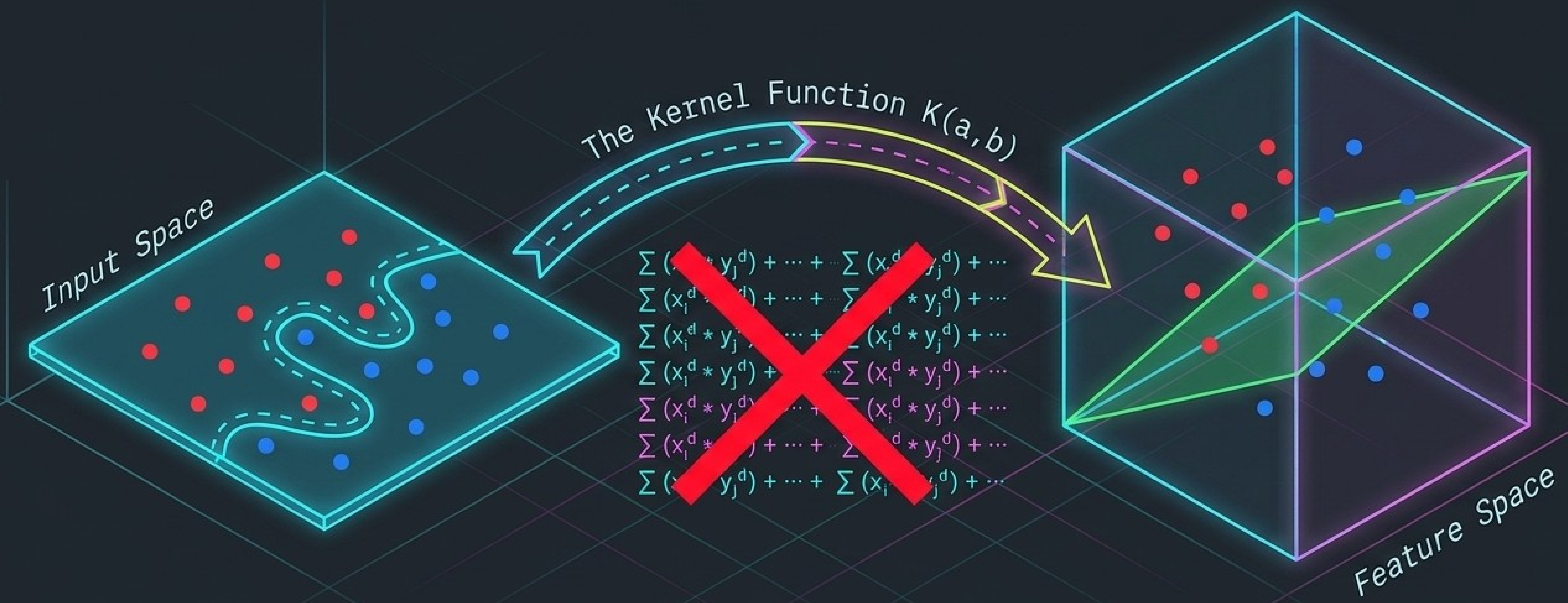
Hard Constraints: Enforces perfect linear separation; highly brittle.
Controlled Violations: Soft margins use slack variables to permit localized errors.
C-Hyperparameter: Tuning dial controlling trade-off between margin width and tolerance.

When Straight Lines Fail



- The Geometric Problem: Complex datasets often cannot be sliced by flat planes.
- Dimensional Evolution: Mathematically map features into a higher-dimensional space.
- The Result: Hopelessly tangled points become perfectly, linearly separable.

Infinite Dimensions, Finite Computation



Combinatorial Explosion:
Computing billions of higher-dimension features crashes hardware.

Mathematical Shortcut:
Kernels calculate spatial relationships exactly as if in higher dimensions.

The Kernel Trick:
Flawless non-linear separation without crippling coordinate transformations.